ment. Most practical robots look like a collection of arms joined together, as pictured. Clearly, this is more like something out of a Mechanix set than what Asimov imagined.

As boring as it may actually sound, practicality is king in the design of almost any robotic body. Often, serious engineering analyses come into the picture. The design team has to account for load-bearing ends, for the structural integrity of the robot, for the available ranges of motion, the speeds that can be sustained - all while trying to minimize the weight, cost, space taken, and complexity of the robot.

In fact, the design of such robots is often classified in an extremely well-defined scientific way. The final arm, where a task is performed, such as holding a camera or picking up an object, is termed as the actuator, while the initial point of reference which remains stationary with respect to the surroundings is called the ground. The ground is connected to the actuator by a series of links. These may be two-dimensional links in simple cases, such as lifting an object and placing it at some fixed distance away. They may also be three-dimensional links, as in the case of a more complicated process of holding an object, rotating it and placing it on a surface after navigating some obstacles. These links are connected to each other by different types of joints. Some of the common types of joints are the prismatic joint (which allows for relative sliding), the rotatory joint (which allows for relative turning), the cylindrical joint (relative sliding and turning simultaneously) and the helical joint (relative sliding and turning bound by a screw-like constraint).

The lengths of different links and their orientations can actually be solved for given the final ranges of motion that the robot intends to possess, this kind of solution typically involves setting up a complicated equation solver over many iterations to find the best possible solution.

How it moves - actuation

A robot might be extremely pretty to look at, or even extremely functional in its range of motion - but it is little more than a hunk of metal until and unless a suitable mechanism is provided to actually set it in motion in a precise and controllable way. This is the field of actuation - literally, the technology behind getting the actuator to go where we want it to!

At the heart of actuation is the use of motors - it is truly amazing what kinds of motion have been derived from the simple rotational motion of a motor. Advances in motor technology have led to innovations like servo-